

Eliminación de Gaussiana

1

$$2x + y = 5$$

$$x + 3y = 6$$

$$\left[\begin{array}{cc|c} 2 & 1 & 5 \\ 1 & 3 & 6 \end{array} \right]$$

$$F_2 = 2F_2 - F_1$$

$$2F_2 = \left[\begin{array}{cc|c} 2 & 6 & 12 \end{array} \right]$$

$$\left[\begin{array}{cc|c} 2 & 6 & 12 \\ 2 & 1 & 5 \end{array} \right] - \left[\begin{array}{cc|c} 2 & 1 & 5 \end{array} \right]$$

$$F_2 = \left[\begin{array}{cc|c} 0 & 5 & 7 \end{array} \right]$$

$$\left[\begin{array}{cc|c} 2 & 1 & 5 \\ 0 & 5 & 7 \end{array} \right]$$

$$2x + y = 5$$

$$5y = 7$$

$$y = \frac{7}{5}$$

$$\boxed{y = 1.4}$$

$$\underline{\underline{x = 1.8 \quad y = 1.4}}$$

$$2x + y = 5$$

$$2x + 1.4 = 5$$

$$2x = 5 - 1.4$$

$$2x = 3.6$$

$$x = \frac{3.6}{2}$$

$$\boxed{x = 1.8}$$

$$\begin{aligned} 2. \quad x + y + z &= 6 \\ 2x + 3y + z &= 14 \end{aligned}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & 3 & 1 & 14 \end{array} \right]$$

$$F_2 = F_2 - 2F_1$$

$$2F_1 = [2 \ 2 \ 2 \ | \ 12]$$

$$\left[\begin{array}{ccc|c} 2 & 3 & 1 & 14 \\ 2 & 2 & 2 & 12 \end{array} \right] - [2 \ 2 \ 2 \ | \ 12]$$

$$F_2 = [0 \ 1 \ -1 \ | \ 2]$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -1 & 2 \end{array} \right]$$

$$\begin{aligned} x + y + z &= 6 & y &= 2 + z \\ y - z &= 2 \end{aligned}$$

$$x + (2 + z) + z = 6$$

$$x + 2 + 2z = 6$$

$$x + 2z = 4$$

$$x = 4 - 2z$$

$$z = 1$$

$$y = 2 + 1 = 3$$

$$x = 4 - 2(1)$$

$$x = 2$$

$$x = 2 \quad y = 3 \quad z = 1$$

$$\begin{aligned} 3. \quad 3x - y + z &= 4 \\ x + 2y - z &= 1 \end{aligned}$$

$$\left[\begin{array}{ccc|c} 3 & -1 & 1 & 4 \\ 1 & 2 & -1 & 1 \end{array} \right]$$

$$F_2 = 3F_2 - F_1$$

$$3F_2 \left[\begin{array}{ccc|c} 3 & 6 & -3 & 3 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 3 & 6 & -3 & 3 \end{array} \right] - \left[\begin{array}{ccc|c} 3 & -1 & 1 & 4 \end{array} \right]$$

$$F_2 = \left[\begin{array}{ccc|c} 0 & 7 & -4 & -1 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 3 & -1 & 1 & 4 \\ 0 & 7 & -4 & -1 \end{array} \right]$$

$$3x - y + z = 4$$

$$7y - 4z = -1$$

$$3x - 0.43 + 1 = 4$$

$$3x + 0.57 = 4$$

$$3x = 3.43$$

$$x = \frac{3.43}{3}$$

$$3$$

$$x \approx 1.14$$

$$z = 1$$

$$7y - 4(1) = -1$$

$$7y - 4 = -1$$

$$7y = 3$$

$$y = \frac{3}{7}$$

$$y \approx 0.43$$

$$y \approx 0.43 \quad z = 1$$

4.

$$4x + 3y = 10$$

$$2x + 5y = 12$$

$$\begin{bmatrix} 4 & 3 & | & 10 \\ 2 & 5 & | & 12 \end{bmatrix}$$

$$F_2 = 2F_2 - F_1$$

$$2F_2 = \begin{bmatrix} 4 & 10 & | & 24 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 10 & | & 24 \end{bmatrix} - \begin{bmatrix} 4 & 3 & | & 10 \end{bmatrix}$$

$$F_2 = \begin{bmatrix} 0 & 7 & | & 14 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 3 & | & 10 \\ 0 & 7 & | & 14 \end{bmatrix}$$

$$4x + 3y = 10$$

$$7y = 14$$

$$y = \frac{14}{7}$$

$$y = 2$$

$$4x + 3(2) = 10$$

$$4x + 6 = 10$$

$$4x = 4$$

$$x = \frac{4}{4}$$

$$x = 1$$

$$\underline{\underline{x = 1 \quad y = 2}}$$

$$2x + y - z = 8$$

$$-3x - y + 2z = -11$$

$$z = -1$$

$$2x + y - (-1) = 8$$

$$2x + y + 1 = 8$$

$$2x + y = 7$$

$$-3x - y + 2(-1) = -11$$

$$-3x - y - 2 = -11$$

$$-3x - y = -9$$

$$(-1)(-3x - y = -9)$$

$$3x + y = 9$$

$$2x + y = 7$$

$$3x + y = 9$$

$$(3x + y) - (2x + y) = 9 - 7$$

$$x = 2$$

$$2x + y = 7$$

$$2(2) + y = 7$$

$$4 + y = 7$$

$$y = 3$$

$$x = 2 \quad y = 3 \quad z = -1$$

Encontrar la inversa de
 $\begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$

$$\begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$$

$$\left[\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 1 & 3 & 0 & 1 \end{array} \right]$$

$$F_1 = \frac{F_1}{2}$$

$$\left[\begin{array}{cc|cc} 1 & 0.5 & 0.5 & 0 \\ 1 & 3 & 0 & 1 \end{array} \right]$$

$$F_2 = F_2 - F_1$$

$$\left[\begin{array}{cc|cc} 1 & 3 & 0 & 1 \\ 1 & 0.5 & 0.5 & 0 \end{array} \right] - \left[\begin{array}{cc|cc} 1 & 0.5 & 0.5 & 0 \end{array} \right]$$

$$F_2 = \left[\begin{array}{cc|cc} 0 & 2.5 & -0.5 & 1 \end{array} \right]$$

$$\left[\begin{array}{cc|cc} 1 & 0.5 & 0.5 & 0 \\ 0 & 2.5 & -0.5 & 1 \end{array} \right]$$

$$F_2 = \frac{F_2}{2.5}$$

$$\left[\begin{array}{cc|cc} 1 & 0.5 & 0.5 & 0 \\ 0 & 1 & -0.2 & 0.4 \end{array} \right]$$

FECHA	DIA	MES	AÑO	FOLIO

$$F_1 = f_1 = 0.5 F_2$$

$$0.5 F_2 = [0 \ 0.5 \mid -0.1 \ 0.2]$$

$$[1 \ 0.5 \mid 0.5 \ 0] - [0 \ 0.5 \mid -0.1 \ 0.2]$$

$$f_1 = [1 \ 0 \mid 0.6 \ -0.2]$$

$$\begin{bmatrix} 1 & 0 & \mid & 0.6 & -0.2 \\ 0 & 1 & \mid & -0.2 & 0.4 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 0.6 & -0.2 \\ -0.2 & 0.4 \end{bmatrix}$$

Método de Jacobi

DÍA

MES

AÑO

$$\begin{aligned}4x - y + z &= 7 \\4x - 8y + z &= -21\end{aligned}$$

$$z = 3$$

$$\begin{aligned}4x - y + 3 &= 7 \\4x - y &= 4\end{aligned}$$

$$\begin{aligned}4x &= 4 + y \\x &= \frac{4 + y}{4}\end{aligned}$$

$$x_0 = 0$$

$$x_1 = \frac{4 + 0}{4}$$

$$x_1 = 1$$

$$x_2 = \frac{4 + 3}{4}$$

$$x_2 = \frac{7}{4}$$

$$x_2 = 1.75$$

$$\begin{aligned}4x - 8y + 3 &= -21 \\4x - 8y &= -24\end{aligned}$$

$$\begin{aligned}-8y &= -24 - 4x \\y &= \frac{-24 - 4x}{-8}\end{aligned}$$

$$y = 3 + \frac{x}{2}$$

$$y_0 = 0$$

$$y_1 = 3 + \frac{0}{2}$$

$$y_1 = 3$$

$$y_2 = 3 + \frac{1}{2}$$

$$y_2 = 3.5$$

FECHA	DIA	MES	AÑO	FOLIO

$$X_3 = \frac{4 + 3.5}{4}$$

$$X_3 = 1.875$$

$$Y_3 = \frac{3 + 1.75}{2}$$

$$Y_3 = 3.875$$

$$X \approx 2$$

$$y \approx 4$$

$$z = 3$$

x

$$10x + y = 12$$

$$x + 10y = 12$$

$$10x + y = 12$$

$$10x = 12 - y$$

$$x = \frac{12 - y}{10}$$

$$x + 10y = 12$$

$$10y = 12 - x$$

$$y = \frac{12 - x}{10}$$

$$x_0 = 0$$

$$y_0 = 0$$

①

$$x_1 = \frac{12 - 0}{10} = 1.2$$

$$y_1 = \frac{12 - 0}{10} = 1.2$$

②

$$x_2 = \frac{12 - 1.2}{10}$$

$$y_2 = \frac{12 - 1.2}{10}$$

$$x_2 = \frac{10.8}{10} = 1.08$$

$$y_2 = \frac{12 - 1.2}{10}$$

$$y_2 = 1.08$$

③

$$x_3 = \frac{12 - 1.08}{10}$$

$$y_3 = \frac{12 - 1.08}{10}$$

$$x_3 = \frac{10.92}{10} = 1.092$$

$$y_3 = 1.092$$

④

$$x_4 = \frac{12 - 1.092}{10}$$

$$y_4 = \frac{12 - 1.092}{10}$$

$$x_4 = \frac{10.908}{10} = 1.0908$$

$$y_4 = 1.0908$$

$$x \approx 1.09$$

$$y \approx 1.09$$

$$\begin{aligned} 5x + y &= 6 \\ x + 5y &= 6 \end{aligned}$$

$$\begin{aligned} 5x + y &= 6 \\ 5x &= 6 - y \\ x &= \frac{6 - y}{5} \end{aligned}$$

$$\begin{aligned} x + 5y &= 6 \\ 5y &= 6 - x \\ y &= \frac{6 - x}{5} \end{aligned}$$

$$x_0 = 0$$

$$y_0 = 6$$

①

$$x_1 = \frac{6 - 0}{5}$$

$$y_1 = \frac{6 - 0}{5}$$

$$x_1 = 1.2$$

$$y_1 = 1.2$$

②

$$x_2 = \frac{6 - 1.2}{5}$$

$$y_2 = \frac{6 - 1.2}{5}$$

$$x_2 = \frac{4.8}{5}$$

$$y_2 = 0.96$$

$$x_2 = 0.96$$

③

$$x_3 = \frac{6 - 0.96}{5}$$

$$y_3 = 6 - 0.96$$

$$y_3 = 1.008$$

$$x_3 = \frac{5.04}{5}$$

$$x_3 = 1.008$$

(4)

$$x_4 = \frac{6 - 1.008}{5}$$

$$x_4 = \frac{4.992}{5}$$

$$x_4 = 0.9984$$

$$y_4 = 6 - 1.008$$

$$y_4 = 0.9984$$

Convergen a

$$x \approx 1 \quad y \approx 1$$

Gauss-Seidel

DÍA

MES

AÑO

FOLIO

$$4x - y + z = 7$$

$$4x - 8y + z = -21$$

$$z = 3$$

$$4x - y = 4$$

$$4x - 8y = -24$$

$$x = \frac{4 + y}{4}$$

$$y = 3 + \frac{x}{2}$$

$$x_0 = 0$$

$$y_0 = 0$$

$$x_1 = 1$$

$$\textcircled{1} \quad x_1 = \frac{4 + 0}{4}$$

$$x_1 = 1$$

$$y_1 = 3 + \frac{1}{2}$$

$$y_1 = 3.5$$

$$\textcircled{2} \quad x_2 = \frac{4 + 3.5}{4}$$

$$x_2 = 1.875$$

$$y_2 = 3 + \frac{1.875}{2}$$

$$y_2 = 3.9375$$

③

$$x_3 = \frac{4 + 3.9375}{4}$$

$$y_3 = 3 + 1.9844$$

$$y_3 = 3.9922$$

$$x_3 = 1.9844$$

$$x \approx 2 \quad y \approx 4 \quad z \approx 3$$

El método de Gauss-Seidel converge más rápido que Jacobi.

$$10x - y + 2z = 6$$

$$-x + 11y - z + 3w = 25$$

$$x = 1$$

$$y = 2$$

$$z = -1$$

$$w = 1$$

$$10x - y + 2(-1) = 6$$

$$10x - y - 2 = 6$$

$$10x - y = 8$$

$$-x + 11y(-1) + 3(1) = 25$$

$$-x + 11y + 1 + 3 = 25$$

$$-x + 11y = 21$$

$$10x - y = 8$$

$$10x = 8 + y$$

$$x = \frac{8 + y}{10}$$

$$-x + 11y = 21$$

$$11y = 21 + x$$

$$y = \frac{21 + x}{11}$$

$$\textcircled{1} \quad x_1 = \frac{8 + 0}{10}$$

$$x_1 = 0.8$$

$$y_1 = 21 + 0.8$$

$$y_1 = \frac{21.8}{11}$$

$$x_1 = 1.9818$$

$$\textcircled{2} \quad x_2 = \frac{8 + 1.9818}{10}$$

$$x_2 = 0.99818$$

$$y_2 = 21 + 0.99818$$

$$y_2 = 1.99983$$

③

$$x_3 = 8 + 1.99983$$

$$x_3 = 0.99998$$

$$y_3 = \frac{21 + 0.99998}{11}$$

$$y_3 = 2.0000$$

Los valores se estabilizan

$$x \approx 1 \quad y \approx 2$$

$$z = -1 \quad w = 1$$

$$x = 1 \quad y = 2 \quad z = -1 \quad w = 1$$